

Knzhou

Handouts

P1 — M2.5

06/10/25
⑨ [4=2] PDT

P1

E1: Find how $v \propto \omega^n$
 $[L][T]^{-1} = [T]^n$

Dep Var:

ω , μ , l

$v \propto \omega$

remark: why no T ?

E2 Find $[B]$

$$F = q \times (v \times B)$$

$$\frac{[M][L][T]^{-2}}{[C][L][T]^{-1}} = B = \frac{kg}{C \cdot s}$$

P1: $P = \frac{dW}{dt} = \frac{[M][L]^2[T]^{-2}}{[T]} = \frac{kg \cdot m^2}{s^3}$

$$G = \frac{m^3}{kg \cdot s^2} = \frac{N \cdot m^2}{kg^2}$$

$$\epsilon_0 = \frac{s^2 C^2}{kg \cdot m^3}$$

$$R = \frac{kg \cdot m^2}{mol \cdot K \cdot s^2} = \frac{N \cdot m}{mol \cdot K}$$

P2

$\alpha \propto r^x$ T could depend on: G, M, r

$$[T] = [G]^x [M]^y [r]^z$$

$$[T] = [L]^x [T]^{-2x} [L]^y [M]^y [L]^z$$

$$x = -\frac{1}{2} \quad y = -\frac{1}{2} \quad z = \frac{3}{2}$$

$$T = r^{\frac{3}{2}} \sqrt{\frac{C}{GM}} \quad T^2 = r^3 \left(\frac{4\pi}{GM} \right)$$

P3 a) $f = R^x \rho^y G^z$

$$[T]^{-1} = [L]^{x-3y+3z} [T]^{-2z} [M]^{y-z}$$

$$z = \frac{1}{2} \quad y = \frac{1}{2} \quad x = 0$$

$$f \propto \sqrt{\rho G} \approx 3 \times 10^{-4} \text{ Hz}$$

$$f \propto \sqrt{\rho R}$$

blind

$$= C \sqrt{\rho R} = C \cdot \sqrt{1410 \text{ kg/m}^3 \cdot 6.963 \times 10^8 \text{ m}}$$

$$= C \cdot 9.901 \times 10^5 \text{ s}^{-1}$$

Remark: $\frac{1}{\sqrt{\rho G}}$ free fall time, is needed

b) $\therefore$ Zero gravity $\therefore G$ doesn't matter. The time needed for a ball of gas of density ρ to collapse is in this order.

$$[T]^{-1} = R^x \rho^y G^z$$

$$= [L]^{x-3y} [T]^{-2z} [M]^{z+y}$$

$$z = \frac{1}{2}, \quad y = -\frac{1}{2}, \quad x = -\frac{3}{2}$$

$$f = C \sqrt{\frac{y}{\rho R^3}}$$

P4 a) independence of $\left\{ \begin{array}{l} T: \text{introduces } k. \\ R: \text{introduces } \text{mol}, \text{ which is unnecessary.} \end{array} \right.$

$$v \propto P^x \rho^y \Rightarrow [L][T]^{-1} = [M]^{x+y} [L]^{-x-3y} [T]^{-2x}$$

$$x = \frac{1}{2} \quad y = -\frac{1}{2} \quad v \propto C \sqrt{\frac{P}{\rho}}$$

$$v = C \cdot \sqrt{\frac{1.01325 \times 10^5 \text{ Pa}}{1.225 \text{ kg/m}^3}} = C \cdot 287.6 \text{ m/s}$$

b) $v \propto B^x \rho^y$

$$[L][T]^{-1} = [M]^{x+y} [L]^{3x-y} [T]^{-2y-2x}$$

$$y = -1 \quad x = -1$$

$$y = \frac{1}{2} \quad x = -\frac{1}{2} ??$$

$$= [L]^{-x-3y} [M]^{x+y} [T]^{-2x}$$

$$x = \frac{1}{2} \quad y = -\frac{1}{2} \quad \text{and } v \propto \sqrt{\frac{B}{\rho}}$$

Remark = think about what the positions mean, given that sound is a pressure wave. blind, didn't saw wavelength

c) When on top, doesn't depend on h .

$$v \propto \rho^x \gamma^y g^z \quad [L][T]^{-1} = [L]^{-3x+z} [M]^{x+y} [T]^{-2y-2z}$$

$$v = c \sqrt{\frac{\gamma g}{\rho}} \quad \text{X} \quad [L][T]^{-1} = [L]^{-3x+z} [M]^{x+y} [T]^{-2y-2z}$$

$x = -y$ $3y+z=1$ $-2y-2z=-1$
 $z = -\frac{1}{2}$ $-2x=1 \Rightarrow x = -\frac{1}{2}$ $y-z=0 \Rightarrow y=z$
 $y = \frac{1}{2}$

d) When very deep, surface tension doesn't matter.

$$v \propto \rho^x h^y g^z = [M]^{x+z} [L]^{-3x+y+z} [T]^{-2z}$$

$$z = \frac{1}{2} \quad x = -\frac{1}{2} \quad y = -1$$

$$v = c \sqrt{\frac{g}{\rho h^2}} \quad \text{X}$$

P5

$$a) \frac{dv}{dt} = -\frac{b}{m} v^n$$

$$\text{when } t=0, v=v_0$$

$$dt = \frac{m}{b} - \frac{1}{v^n} dv$$

$$t = \frac{m}{-b} \left\{ \begin{array}{ll} \log v & \text{if } n=1 \\ \frac{-n+1}{v^{n-1}} & \text{if } n > 1 \\ v & \text{if } n=0 \end{array} \right.$$

Tell what to say I didn't use dim ana

When $n=0$, $v = -\frac{b}{m}t + v_0$
 $t_m = \frac{m}{b}v_0$
 $x_m = \frac{1}{2} \frac{m}{b} v_0^2$

When $n=1$, $v = e^{-\frac{b}{m}t} + (v_0 - 1)$
 $t_m = \infty$
 $x_m = \infty$

When $n=2$, $v = \frac{m}{b} \frac{1}{t + \frac{m}{b}v_0}$
 $t_m = \infty$
 $x_m = \frac{m}{b} \ln \left(\frac{b}{m}v_0 + 1 \right)$

When $n=3$, $v = \sqrt{\frac{m}{2b \left(t + \frac{m}{2b}v_0 \right)}}$

$t_m = \infty$

$x_m = \frac{1}{2} \frac{m}{2b} v_0$

P6 $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, find $\int_{-\infty}^{\infty} e^{-ax^2+bx+c} dx$ -2

• Un-dimensionalize $-ax^2+bx+c \rightarrow -b^2/a$

let $u = \frac{b}{\sqrt{a}}x$

Can't do independently ...

• Separate useless constants

• 配方 + substitute 消一次项

$= e^c \int_{-\infty}^{\infty} e^{-ax^2+bx} dx$

$x = \frac{1}{b\sqrt{a}}u$
 Sub $\rightarrow e^{-\frac{1}{b^2}u^2 + \frac{u}{\sqrt{a}}}$

P7. $125 \text{ AeV} \rightarrow \frac{125 \text{ AeV}}{(\text{light second/second})^2}$
 $= 2.22 \times 10^{-22} \text{ kg} \leftarrow \text{g}$
 $= 2.22 \times 10^{-25} \text{ kg}$

P8. 自己做不出, 当 insight 理解

We have a ∇^2 in the KE term, which gives ψ an additional dimension of $[L]^{-2}$, thus the equation can be rearranged as,

$$\frac{\hbar^2}{2mr^2} \sim \frac{e^2}{4\pi\epsilon_0 r}$$

where r is the distance of the electron to the nucleus. we have,

$r \sim a$
 where a is the radius of the hydrogen atom, therefore

$$\frac{\hbar^2}{2ma^2} \sim \frac{e^2}{4\pi\epsilon_0 a}$$

Solving for a yields that

$$a \sim \frac{4\pi\epsilon_0 \hbar^2}{2me^2} \sim 10^{-10} \text{ m}$$

P10

$$a) E \propto R^a t^b \rho^c$$

$$[M][L]^2[T]^{-2} \propto [M]^c [L]^a [T]^b$$

$$E \propto \frac{R^5}{t^2} \approx \cancel{4.44 \times 10^7} = 4.44 \times 10^{13} \text{ J}$$

→ 0.5
(wrong calculation)

$$b) m = \frac{E}{c^2} = \cancel{4.94 \times 10^{-10}} \text{ g} = \cancel{4.44} 0.44 \text{ g}$$

$$c) t^2 \propto R^5 \quad R \propto t^{\frac{2}{5}}$$

$$a_0 = \frac{h}{mv}$$

$$P11 \quad a) \hat{\nabla} = a_0 \nabla \quad a_0 \nabla = a_0 \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)$$

$$\frac{d\psi}{dx} = \frac{d\psi}{dx} \quad \frac{d\psi}{dx} = \frac{d\psi}{dx} \frac{dx}{dx} \quad \bar{\psi}$$

$$b) -\frac{\hbar^2}{2m} \nabla^2 \psi - \frac{e^2}{4\pi\epsilon_0 r} \psi = E \psi$$

$$\hbar \text{ has dim of } ML^2T^{-1} \quad m \text{ has } M \quad \frac{e^2}{4\pi\epsilon_0} \text{ has } ML^3T^{-2}$$

$$E \text{ has } ML^2T^{-2}$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi$$